

Requirement Analysis Phase

Solution Requirements

Date	25 th July 2026
Project Name	Metro Ticket Booking System

Solution Requirements

Introduction

The Metro Ticket Generation System is designed to automate the metro ticket booking process using ServiceNow. The system enables passengers to select journey details, automatically calculates fares, generates QR-based digital tickets, and provides instant notifications. The solution aims to reduce manual effort, minimize waiting time, and improve the overall commuter experience.

Functional Requirements

Passenger Requirements

- User can access the Metro Ticket Booking Catalog.
- User can select source and destination stations.
- User can choose journey type (Single/Return).
- User can enter the number of passengers.
- User can select the travel date.
- System validates all mandatory fields.
- System automatically calculates the fare.
- User can choose the ticket generation method (QR Ticket/Manual Ticket).
- System generates a unique ticket number.
- System generates a QR code for the booked ticket.
- User receives booking confirmation through email or notification.

- User can view booking history.

Admin Requirements

- Add/Edit/Delete metro stations.
- Configure fare chart.
- View passenger bookings.
- Cancel bookings if required.
- Monitor ticket statistics.
- Generate reports.
- Manage QR ticket verification.

Ticket Inspector Requirements

- Scan QR Code.
- Verify ticket validity.
- Check journey details.
- Mark ticket as used.
- Reject invalid or expired tickets.

Non-Functional Requirements

Performance

- Ticket generation within 2–3 seconds.
- Fast fare calculation.

Security

- Role-based access control (RBAC).
- Secure ticket storage.
- QR code validation.

Reliability

- 24/7 system availability.
- Consistent ticket generation.

Scalability

- Support thousands of daily bookings.
- Future integration with payment gateways.

Usability

- Simple Service Portal interface.
- Mobile-friendly booking experience.

Assumptions

- Passenger has internet access.
- Metro stations are pre-configured.
- Fare data is maintained by administrators.
- ServiceNow instance is available.

Constraints

- Depends on ServiceNow platform.
- QR verification requires scanner support.
- Internet connectivity required.